

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

PHASE 22

THE QUINN STORAGE TANKS. BABUSHKA NESTED 2300 PSI SCISSOR-BRACE GAS STORAGE TANKS

Ten bullet tanks nested — sacrificial sleeves — spring-tension yokes
Launch-and-overtake: the highway's own supply chain



SCISSOR BRACES — LOADS ROUTE AROUND

NESTED EMPTY — FILLED AT DEPLOYMENT — NEVER STORED FULL

HOOP STRESS $\sigma = P \cdot r / t$ — 2300 psi IS A DESIGN PRESSURE, NOT GEOMETRY MAGIC

GP-IM-022

Revision v1.2 — 21 August 2026

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VETTED: PASS — HULL-GRADE SHELLS, NO ESCORT

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LICENSE: CERN OPEN HARDWARE LICENCE v2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — CHAPTER 22

PHASE 22: THE QUINN STORAGE TANKS. BABUSHKA NESTED 2300 PSI SCISSOR-BRACE GAS STORAGE TANKS — STATUS: CURRENT. The ship's nitrogen vault: ten bullet-shaped high-pressure tanks nested inside one another — scissor-brace cross-members that fold flat to load, then spring-drop into rigid load yokes; sacrificial polymer sleeves between skins so no metal ever scores metal. The architecture's value is honest: one puncture is not one catastrophe — an outer vessel takes the hit, the inner holds the gas. And the logistics half: house-sized tanks launched ahead at 5,000 mph, the ship at 30,000 mph overtaking and capturing them in the gasless voids. Vetted: PASS architecture / pressure-vessel certification required — the hoop-stress law is seated in §2. [CORRECTED 21 AUGUST 2026 — his ruling: the 'one puncture' value was a MISREAD (two launched at once, in case one got hit); there is NO puncture containment; the tanks nest EMPTY and are never stored full — §1 governs.]

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-022, v1.2 — 21 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the Babushka law as seated (contents line and design bodies, verbatim) — §2 why it works (nesting physics, graded; the hoop-stress discipline) — §3 the system on the ship — §4 the doctrine record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Rejects explosive, single-wall fuel reservoirs. Implements a multi-layered, nested tank geometry held under 2300 Psi, locked within high-torque mechanical scissor-braces to isolate and absorb severe structural hull impacts. For nitrogen collection for prilling and hydrogen of ship storage.'

THE SEATED DESIGN BODY — THE CELLS (verbatim): 'Babushka Storage Cells: 10 bullet-shaped high-pressure (2300 psi) nitrogen supply tanks nested inside each other. Internal structural cross-braces scissor flat to load a tank, then drop via spring-tension into rigid load yokes. High-density polymer sacrificial buffer sleeves insulate the skins against metal scoring.'

THE SEATED DESIGN BODY — THE LOOP (verbatim): 'Launch-and-Overtake Loop: House-sized Babushka tanks are launched ahead of the vessel at low speed (e.g., 5,000 mph). The starship cruising at 30,000 mph overtakes and captures the pods deep within gasless voids.'

THE NITROGEN BASELINE (verbatim, the phase's own block — July transcript truncations preserved): 'Biological Imperative: Rejects standard low-barometric space atmosphere to prevent deep-voyage human musculoskeletal circulatory degradation. Maintaining an Earth 14.7 psi (1 bar) profile requiring continuous Nitrogen volumes. Sourcing Matrix: Automated lower-deck retrieval craft execute upper skimming sweeps across nitrogen-dense planetary atm[ospheres —] filtering and compressing the raw gas into the engi[neering stores — the transcript's own cut, preserved].' THE 2300 PSI CELL, same block (verbatim): 'Structural Geometry: 10 heavy, bullet-shaped armored tanks nested inside one another to minimize storage space. Internal Reinforcement: Integrated geodesic interior space-frame mat[rix bracing —] the structural walls against sustained 2,300 [psi —]. Interlocking Mechanism: High-torque screw-seal locking pins anchoring concentric layer[s].'

THE INVENTOR'S PREFERENCE (verbatim, same block): 'Rear launch and catchup methodologies following the ship can also be employed (inventor preference, due to uncleared forward path.)' — and the redundant contingency: 'Deploys dual parallel supply cells through hi[gh-risk —] or barren void sectors to guarantee retention in the event of a micro-meteorite strike taking out the supply.' [NOTE, 21 AUGUST 2026 — THIS dual-cell redundancy is the true source of the puncture-survival memory: two filled cells launched at the same time, in case one got hit. The nesting was never the protection.]

THE EMPTY-NEST LAW AND THE SEQUENCE (his ruling, verbatim): "empty tanks sit inside each other, smallest out first, base sealed, filled to pressure with the required gas made on board and launched, there is never a tank inside a tank when filled, the error is saying if one is punctured the other survives, this was misread from 2 being launched at the same time in case one got hit"

disc banks no longer exist so remove that entirely

build sequence is wrong

babushka, large tank fold back scissor brace, insert protection sleeve , insert next smaller tank, repeat.

there is no puncture containment for babushkas, they are only filled before deployment never stored

full on the ship. the ship of course has tanks, these are like Jerry cans not kept on board"

SEATED AS THE EMPTY-NEST LAW (audit reading, disclosed): (1) THE NEST IS A STORAGE GEOMETRY, NEVER A FILLED STATE — empty tanks sit inside each other; there is NEVER a tank inside a tank when filled. (2) DEPLOYMENT — smallest out first, base sealed, filled to pressure with the required gas MADE ON BOARD, and launched (ahead-loop or rear-launch catchup, his standing preference). (3) THE MISREAD, OWNED BY THE AUDIT TOO — 'if one is punctured the other survives' was never the architecture: it was misread from the dual-launch redundancy — TWO filled tanks launched at the same time in case one got hit. There is NO puncture containment for babushkas. (4) THE JERRY-CAN DOCTRINE — they are filled only before deployment, never stored full on the ship; the ship of course has its own tanks — these are like Jerry cans, not kept on board. (5) THE BUILD SEQUENCE, CORRECTED on his order ('build sequence is wrong') — the chapter's innermost-first sequence was backwards: his sequence is large tank, fold back scissor brace, insert protection sleeve, insert next smaller tank, repeat — the nest assembles INWARD from the largest. (6) KILL ORDER EXECUTED — 'disc banks no longer exist so remove that entirely': the escort doctrine and the Phase 160 cross-reference are removed from this chapter ENTIRELY (the removed texts are archived verbatim in SITTING 515); the master's archive carries kill brackets on the old lines per verbatim-preservation law.

THE SHELLS-ONLY LAW AND THE WINGS BAR (his ruling, verbatim): "the disc protection was when we could move small numbers in banks, now it isn't feasible, we cannot use the wings as protection as they must remain protection for the ship and has become our main generator, a blast by a puncture between the wing and ship would be a disaster, the gas dolls are the same hull as the ship pretty much, so it is what it is, we cannot build and carry huge equipment for everything. the babushka doll shells are their only protection now"

SEATED AS THE SHELLS-ONLY LAW (audit reading, disclosed): (1) THE DISC ESCORT IS GONE WITH ITS LOGISTICS — the disc banks existed to protect small numbers of pods moving in banks; banked movement is no longer feasible, so the escort has no mission. (2) THE WINGS BAR — the wings cannot be reassigned as pod protection: they must remain protection for the ship, and they have become our main generator; and never station pods in the wing-ship gap — a puncture blast there would be a disaster. (3) HULL-GRADE SHELLS — the gas dolls are 'the same hull as the ship pretty much'; the babushka doll shells are their only protection now. (4) THE NO-HUGE-EQUIPMENT DOCTRINE — 'it is what it is, we cannot build and carry huge equipment for everything.' Disclosed audit note: 'the wings' is read as the bay eaves named 'wings' (master), whose seated duty is deflection-first protection of the ship; the main-generator duty on them is new law by his word, and its spec rides §9.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE NESTING PHYSICS: there is nothing physically impossible about nesting pressure vessels — impact

protection, redundancy, shielding, separation, puncture resistance, controlled failure (the vet's list). The scissor-brace structure transfers external loads AROUND the pressure vessel rather than letting a local impact load the skin directly.

THE HOOP-STRESS DISCIPLINE, SEATED AS LAW OF READING: '2300 psi held by geometry' does NOT survive — geometry does not magically establish a pressure rating. $\sigma_h = P \cdot r / t$: capacity depends on material, wall thickness, radius, stress concentration, weld/joint design, fatigue, temperature, safety factor, pressure-cycle count. 2,300 psi is a DESIGN PRESSURE — engineered and certified, never inherited from the nesting.

THE ONE-PUNCTURE VALUE, STATED HONESTLY: if an outer vessel is punctured, the inner can remain intact — that is the architecture's actual value. It does NOT mean 'one puncture is never one catastrophe' as an absolute guarantee — the caution stands; the book's own summary line (§4) reads as shorthand, governed by this paragraph. [OVERTURNED 21 AUGUST 2026 — by its author: there is NO puncture containment for babushkas; nested tanks are never filled; the value claim above is dead and stands only as the record of the misread.]

THE LOGISTICS PHYSICS: launch-and-overtake works because the void is gasless — no drag differential to erase the geometry. The pod ahead at 5,000 mph and the ship at 30,000 mph share one vector; the overtake is a capture problem the Phase 20 tether discipline already owns (guide, not mitt — match before the line goes taut). The inventor's stated preference — rear launch and catchup, the uncleared forward path — is seated with it.

§3 — THE SYSTEM ON THE SHIP

- THE TEN: bullet-shaped armored tanks, concentric, nitrogen at the 2,300 psi design pressure — the Earth-normal 14.7 psi cabin profile's reserve vault. [CORRECTED 21 AUGUST 2026 — the ten ride aboard EMPTY, nested for storage volume only; filled only at deployment, never stored full.]
- THE BRACES: cross-members scissor flat to load a tank, then spring-tension drop into rigid load yokes; high-torque screw-seal locking pins anchor the concentric layers.
- THE SLEEVES: high-density polymer sacrificial buffers between skins — the nesting never trades metal scoring for impact safety.
- THE EMPTY NEST (his law, 21 August 2026): the ten ride aboard EMPTY — nesting is storage volume, never a filled state; there is never a tank inside a tank when filled.
- THE JERRY-CAN DOCTRINE (his words, seated): filled only before deployment, never stored full on the ship — 'the ship of course has tanks, these are like Jerry cans not kept on board.' The ship's own fixed tanks are separate architecture.
- THE SHELLS ARE THE PROTECTION (his law, 21 August 2026): 'the babushka doll shells are their only

protection now' — the dolls are 'the same hull as the ship pretty much': hull-grade skins, no escort, no shield barge, no disc bank. 'It is what it is, we cannot build and carry huge equipment for everything.'

- THE WINGS BAR (his law): the wings are NOT pod protection — they must remain protection for the ship and have become the main generator; never station pods in the wing-ship gap — 'a blast by a puncture between the wing and ship would be a disaster.'
- THE LOOP: house-sized autonomous supply nodes launched along the future vector (or rear-launched, his preference); onboard computers map consumption against momentum and schedule the overtake to the exact depletion mark; dual parallel cells through high-risk or barren sectors.
- THE SKIMMERS: automated lower-deck retrieval craft skim nitrogen-dense upper atmospheres, filtering and compressing raw gas into the engineering stores.

INTEGRATION NOTE (audit, flagged): the tank farm is mass and hazard — placement, blast isolation and the retrieval craft's bay ride the hull architecture (GP-IM-006 v1.4); the rotational track-ring mounting (§4) seats in the same architecture. Certification figures ride §9.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE ROTATIONAL EQUILIBRIUM : 'Inner fuel cells of the Phase 22 Babushka tanks and all operational tanks are mounted on low-power rotating track rings to override zero-G fluid floating' — slow-spin rotation pins the liquid mass against the perimeter walls: centrifugal containment, settled fluids, no float.
- THE BOOK'S OWN SUMMARY (governed by §2's discipline): 'Phase 22 — Babushka Nested Gas Tanks: tanks inside tanks with scissor braces — 2300 psi held by geometry, so one puncture is never one catastrophe.' — shorthand; the certified reading is §2: 2,300 psi is a design pressure, and the puncture value is protection, not guarantee. [OVERTURNED 21 AUGUST 2026 — the summary's 'one puncture is never one catastrophe' is overturned by its author; the line stays as archive only, no longer shorthand for anything.]
- THE PRILLING LINE (seated in the contents line itself): the nitrogen feeds 'prilling and hydrogen of ship storage' — the vault feeds the materials line, not just the cabins.

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- STEP 1 — THE LARGE TANK: build and certify the largest tank as a pressure vessel (material, wall,

radius, welds, fatigue, cycle count to the 2,300 psi design pressure with its safety factor) — and seat it as the nest's BASE. Base sealed.

- STEP 2 — FOLD BACK THE SCISSOR BRACE: the cross-members scissor flat, opening the nest for loading.
- STEP 3 — INSERT THE PROTECTION SLEEVE: the high-density polymer sacrificial buffer seats against the large tank's inner skin.
- STEP 4 — INSERT THE NEXT SMALLER TANK.
- STEP 5 — REPEAT — brace folds, sleeve, next smaller tank — down to the smallest. The nest assembles INWARD from the largest; his sequence, verbatim: 'babushka, large tank fold back scissor brace, insert protection sleeve , insert next smaller tank, repeat.'
- STEP 6 — THE SET RIDES EMPTY. No tank is ever filled while nested; nothing is stored full on the ship. At deployment: smallest out first, base sealed, fill to pressure with the required gas made on board, and launch — ahead-loop or rear-launch catchup (his preference, seated §1).
- STEP 7 — COMMISSION THE LOOP: the launch/overtake scheduling computer, consumption-vs-momentum mapping, capture integration with the Phase 20 tether discipline — then the certification bench (§9): proof, burst, cycle life. No puncture-containment bench exists — there is no puncture containment.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 20 — the space-anchor tugs (GP-IM-020 v1.1): the overtake-and-capture discipline the supply loop inherits — match speed, guide the line, never mitt the catch.
- PHASE 13 — the 94mm stack (GP-IM-013 v1.4): the armored skins' plating reference; the farm's bay is hull furniture.
- PHASE 6 — Hull & Structure (GP-IM-006 v1.4): tank-farm placement, the track rings' foundations, the retrieval craft's bay.
- PHASE 9 — the Coldrock insulation: the cluster mate — thermal discipline around the vault rides the same architecture.
- PHASE 28/29 — the shuttle & launcher cluster mates: the vacuum-coring and Mayernick launch bodies sit with them in the seated record (§1).

§8 — DO-NOT-BUILD

- NEVER say 'held by geometry' — 2,300 psi is a design pressure, certified per vessel; the hoop-stress

law is seated (§2).

- NEVER let skins touch metal-to-metal — the sacrificial sleeve is the architecture, not an accessory (§1, §6).
- NEVER fly a single supply cell through high-risk or barren sectors — dual parallel cells, seated doctrine (§1).
- NEVER load a brace under tension-side shortcut — scissor flat to load, spring-drop to yoke; the load path routes around the vessel (§2).
- NO puncture-survival language AT ALL: there is no puncture containment; tanks nest empty, fill at deployment (§2, §5).
- NEVER revive the disc-bank escort — the discs existed to protect small numbers moving in banks; banked movement is no longer feasible, the escort has no mission (§1).
- NEVER use the wings as pod protection or park pods between wing and ship — the wings protect the ship and carry the main generator; a puncture blast in the gap is a disaster (§3).

§9 — OWED

THE BENCH, OWED: the pressure-vessel certification set (proof, burst, fatigue cycle count, weld/joint factors at 2,300 psi design); scissor-brace load-path numbers; sleeve material spec vs thermal/vibration cycle; track-ring spin rates and fluid-settling verification; the loop's scheduling tolerance (consumption map vs overtake window); skimmer harvest rates per atmospheric sweep; the fill-at-deployment rig figures; the smallest-out-first extraction mechanics; the base-seal spec; the wings' main-generator duty — what they carry, capacity, and routing ride the power architecture (§3). All owed items ride the bench until spoken.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-022, v1.2 — CURRENT, seated law, master v2.1 (numerical print order). Built 20 August 2026

PROVENANCE — THE STANDING ORDER, VERBATIM (seated in sitting 499): '16 and then bed for me, after we finish that you can do 17 to 27 and i will read all 10 first thing to see if we can get through them faster in blocks'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart. Next version triggers: the §9 certifications, tank-farm placement rulings, or his word.

END OF CHAPTER 22